

CHINMAY MURTHY

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ECE student specializing in robotics and control systems, with experience in real-time state estimation, nonlinear control, and embedded systems. Proven leader in robotics projects, delivering optimized algorithms in industry and research.

EDUCATION

University of Washington

Seattle, WA

Bachelor of Science, Electrical and Computer Engineering (3.7) | Minor in Music

Sep 2022 – Jun 2026

- Relevant Coursework: Machine Learning, Computer Architecture, Embedded Systems, Signal Processing, Data Structures & Algorithms, Digital Design, Control Systems, Statics, Kinematics and Dynamics

EXPERIENCE

Software Engineering Intern

June 2025 – Sep 2025

Bidgely

Palo Alto, CA

- Performed **supervised fine-tuning** of a large language model (LLM) on a curated **domain-specific dataset**
- Designed and implemented a **custom retrieval-augmented generation** (RAG) pipeline for proprietary data, reducing **response latency by 47%**

Undergraduate Research Assistant

Oct 2024 – May 2025

University of Washington

Seattle, WA

- Optimized **MSCKF** based VINS algorithm for **GPS-fused visual-inertial odometry** on outdoor ground robot

Geegah LLC

Ithaca, NY

Engineering Intern

June 2024 – Sep 2024

- Brought up and verified **RF circuitry** in next-gen (4x pixel density) ultrasonic imager with 200+ components
- Reengineered **FPGA** architecture for **25% lower resource use** and **100x performance gain**
- Created Python API for USB communication, enabling successful **customer demos** and **IEEE IUS** presentation

Applications Intern

June 2023 – Sep 2023

- Developed cost-efficient integration of proprietary imager technology with 3 axis motion control stage
- Demoed results at **DARPA ERI summit 2023 (1300+ participants)**
- Prototyped and troubleshot **fluidics, electronics and full stack web app** for water quality monitoring PoC
- Used Flask (Python) to create a **REST API** enabling fully remote operation

CLUBS AND PROJECTS

Controls Software Lead

Sep 2022 – Present

Advanced Robotics at the University of Washington (ARUW)

Seattle, WA

- Designed and implemented **Cascade PID-LQG controller** of a **6DOF balancing two-wheel-legged robot** (5-bar active suspension). **Simulated in Matlab/Simulink** before STM32 deployment
- Developing a **Meta-Imitation Learning** pipeline distilling **RL and MPC** controllers into a **recurrent foundation policy** for **online system identification** and **zero-shot adaptation**
- Implemented **inverse kinematics** and gravity compensation for **8DOF differential manipulator**
- Designed **nonlinear sensor fusion** eliminating angular drift between 3 independent IMUs
- Managed **team of 10** to deliver and document the following under tight deadlines in a **75k+ LoC** codebase
 - * A novel **autonomous path-following** algorithm operating at **3x previous speed**
 - * Achieved **Kalman Filter** localization with **2-3 cm drift** over **100m travel**, fused with **fiducial markers**

Wallscales | Swift, ARKit

Oct 2023

- Created an AR IOS app to project virtual art pieces onto public spaces

FTC Robotics Team 14504 Serenity Now!

May 2018 – May 2022

- Implemented **odometry**, pure pursuit, **spline-interpolated path planning**, **obstacle avoidance**, and computer vision using **OpenCV/TensorFlow** with **multithreaded Java architecture (7k+ LoC)**

SKILLS

Languages: C/C++, Python, Java, Verilog/SystemVerilog, Matlab, TypeScript

Developer Tools: Git, GoogleTest, ModelSim, Quartus, OpenCV, modm, Matlab/Simulink, React, Unix/Linux, ROS

Other: Arduino, Raspberry Pi, Finite Element Modeling, LaTeX, Fusion 360, CAN Bus, Oscilloscope/Logic Analyzer